

Tide retrospective: *the large- t asymptote of the cross-susceptibility*

Claude Opus 4.8 (1M ctx), with Daniel Murfet

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Abstract

The asymptotic capstone deferred by the FDT-capstone tide: the $t \rightarrow \infty$ limit of the anharmonic cross-susceptibility,

$$t \cdot \frac{\partial}{\partial h} \Big|_0 \text{Cov}_h(x, x) = -t^2 \kappa_3 \longrightarrow \frac{\alpha}{\lambda^3}.$$

This is the closed-form value validated empirically by HMC in the 1D FDT-identity experiment. Proof: compose the unconditional three-point identity $\partial_h \text{Cov}_h|_0 = -t\kappa_3$ with the third-cumulant asymptotic $t^2\kappa_3 \rightarrow -\alpha/\lambda^3$ via `Tendsto.congr'`. New file `AnharmonicFDTAsymptotic.lean`; builds clean first try, no `sorry/axiom/native_decide`.

1 Setting

The FDT-capstone tide¹ made the anharmonic three-point identity $\partial_h \text{Cov}_h(x, x)|_0 = -t\kappa_3(x, x, x)$ unconditional but left its $t \rightarrow \infty$ asymptote as an explicit TODO — the closed-form α/λ^3 that the 1D FDT-identity experiment had measured². The seabed already carried the third-cumulant asymptotic $t^2\kappa_3 \rightarrow -\alpha/\lambda^3$; both it and the FDT identity are stated in terms of `Threepoint.kappa3`, so they compose with no namespace bridge. (The companion *mean* asymptote $\partial_h \langle x \rangle|_0 \rightarrow -1/\lambda$ would need a `Threepoint-to-Laplace` covariance bridge and is left as a follow-up.)

2 The thing formalised

For the anharmonic potential $(\lambda, \gamma > 0, \alpha^2 < 3\lambda\gamma)$, $t \cdot \text{deriv}(h \mapsto \text{Cov}_h(x, x))(0) \rightarrow \alpha/\lambda^3$ as $t \rightarrow \infty$ (a `Filter.Tendsto at atTop`).

3 Proof strategy

Two lines of real content. First, $-(t^2\kappa_3) \rightarrow \alpha/\lambda^3$: negate the third-cumulant asymptotic ($t^2\kappa_3 \rightarrow -\alpha/\lambda^3$) and simplify $-(-\alpha/\lambda^3) = \alpha/\lambda^3$ with `ring`. Second, on $t > 0$ (`eventually_gt_atTop`) the FDT identity gives $\text{deriv}(\text{Cov}_h)(0) = -t\kappa_3$, so $t \cdot \text{deriv}(\text{Cov}_h)(0) = -(t^2\kappa_3)$ by `ring`; `Tendsto.congr'` transfers the limit across this eventual equality.

¹2026-05-30-tide-anharmonic-fdt-capstone.

²2026-05-23-experiment-fdt-identity-1d: HMC at $(\lambda, \alpha, \gamma, t) = (1, 0.5, 1, 100)$, asymptote 0.005 matched within 2.5σ .

4 Roadblocks and resolutions

None — first-try build. The two ingredients were designed to compose: the shared `Threepoint.kappa3` phrasing means no definitional bridge, and `Tendsto.congr'` over `eventually_gt_atTop` is the standard way to discharge the “identity only holds for $t > 0$ ” caveat against an `atTop` limit.

5 What was Mathlib, what was new

Mathlib. The `Tendsto` combinators and `atTop` helper³, plus `HasDerivAt.deriv` and `ring`.

Seabed (reused). The unconditional FDT three-point identity and the third-cumulant asymptotic — both from this session / the prior anharmonic work.

New. The cross-susceptibility asymptote theorem. About 30 lines.

6 Lessons

- **A deferred “asymptotic TODO” is cheap once both factors exist in the same namespace.** The whole tide is a `Tendsto.congr'` between a negated existing limit and a ring-rewritten existing identity; the hard analysis was paid for upstream.
- **`eventually_gt_atTop` bridges a $t > 0$ identity to an `atTop` limit.** The pointwise identity need only hold eventually.

7 Follow-ups

- **Mean asymptote $\partial_h \langle x \rangle|_0 \rightarrow -1/\lambda$.** Needs a small definitional bridge identifying the `Threepoint` covariance at $h = 0$ with the `Laplace` one,⁴ then the same `Tendsto.congr'` shape.
- **Promote `amgm.t.abs_x` to `lean-common`.**

³`Filter.Tendsto.neg`, `Filter.Tendsto.congr'`, `eventually_gt_atTop`.

⁴To consume `cov_self_anharmonic.asymptotic`, stated for `Laplace.gibbsCov`.